Estructuras de Datos no Lineales

2.1. Recorridos de grafos

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Versión 1.0

Recorridos de grafos

Recorrido en profundidad

- Generalización del recorrido en preorden de árboles.

```
#include "listaenla.h"
enum visitas {NO_VISITADO, VISITADO};

Lista<Grafo::vertice> Profundidad(const Grafo& G, Grafo::vertice v)
{
   typedef Grafo::vertice vertice;
   const size_t n = G.numVert();
   vector<visitas> marcas(n, NO_VISITADO);
   Lista<vertice> Lv;
   vertice i = v;
   do {
      if (marcas[i] == NO_VISITADO)
         Lv += Profun(G, i, marcas);
      i = (i+1) % n;
   } while (i != v);
   return Lv;
}
```

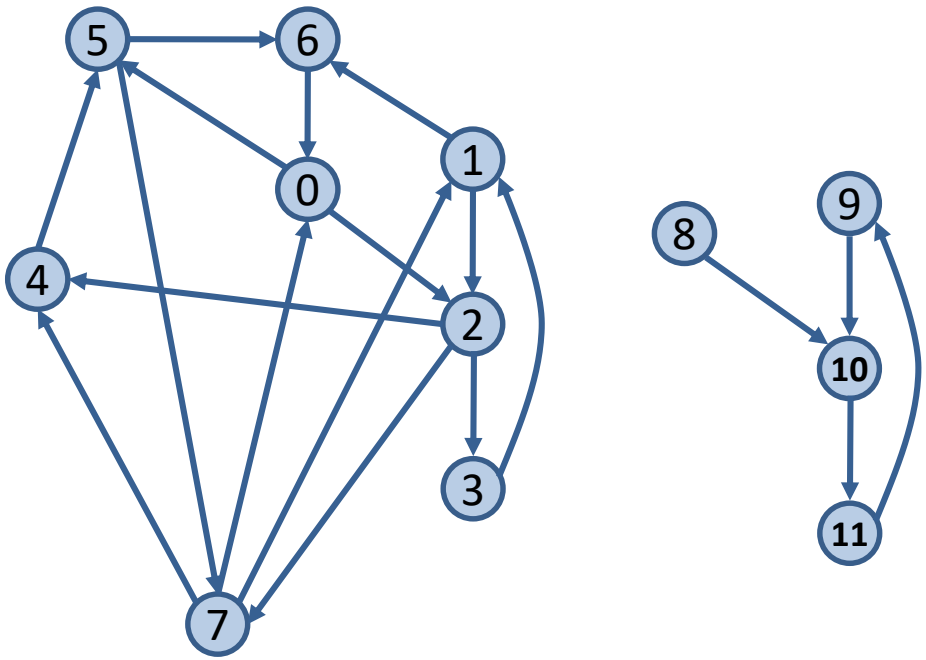
```
static Lista<Grafo::vertice> Profun(const Grafo& G, vertice v,
                                    vector<visitas>& marcas)
{
   typedef Grafo::vertice vertice;
   const size_t n = G.numVert();
   Lista<vertice> Lv;

   marcas[v] = VISITADO;
   Lv.insertar(v, Lv.fin());
   for (vertice w = 0; w < n; w++)
      if (G[v][w] && marcas[w] == NO_VISITADO)
            Lv += Profun(G, w, marcas);
   return Lv;
}
```

```
#include "listaenla.h"
#include "pilaenla.h"
Lista<Grafo::vertice> Profundidad2(const Grafo& G, Grafo::vertice u)
{
   typedef Grafo::vertice vertice;
   const size_t n = G.numVert();
   vector<visitas> marcas(n, NO_VISITADO);
   Pila<vertice> P;
   Lista<vertice> Lv;
   vertice i = u;
   do {
      if (marcas[i] == NO_VISITADO) {
         P.push(i);
         do {
            vertice v = P.tope(); P.pop();
            if (marcas[v] == NO_VISITADO) {
               marcas[v] = VISITADO;
               Lv.insertar(v, Lv.fin());
               // Meter en la pila los adyacentes no visitados
               for (vertice w = n; w > 0; w--)
                  if (G[v][w-1] && marcas[w-1] == NO_VISITADO)
                     P.push(w-1);
            }
         } while (!P.vacia());
      }
      i = (i+1) % n;
   } while (i != u);
   return Lv;
}
```

Ejemplo de recorrido en profundidad

i = 0

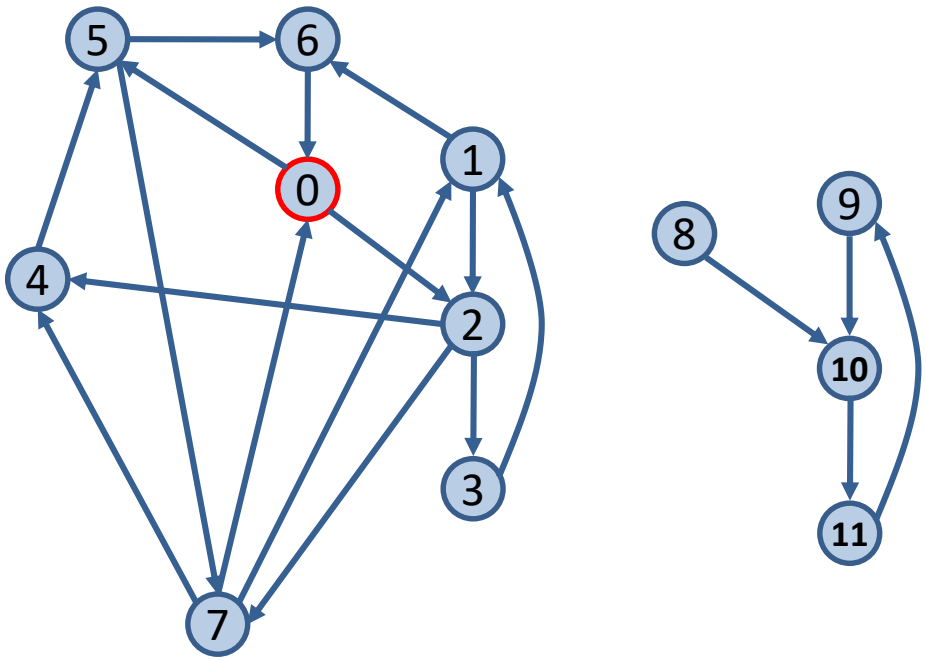


P = {0}

Lv = { }

Ejemplo de recorrido en profundidad

i = 0

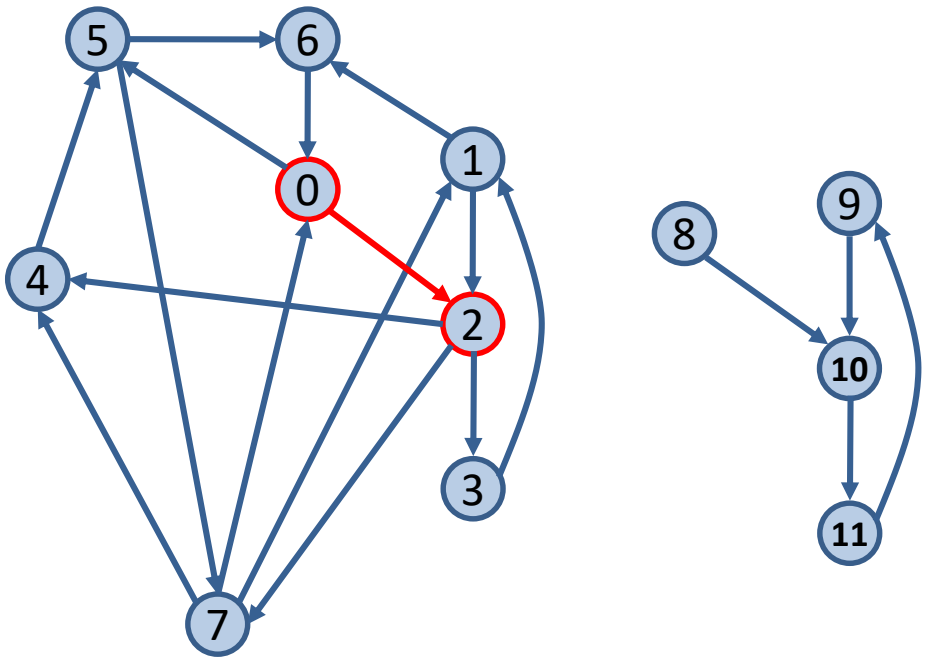


P = {2, 5}

Lv = {0}

Ejemplo de recorrido en profundidad

i = 0

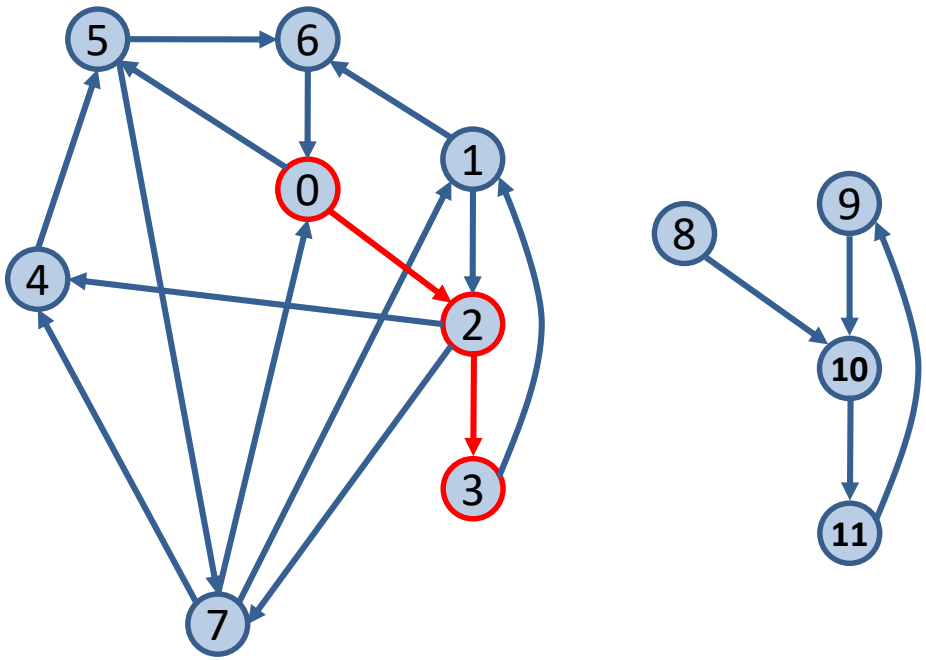


P = {3, 4, 7, 5}

Lv = {0, 2}

Ejemplo de recorrido en profundidad

i = 0

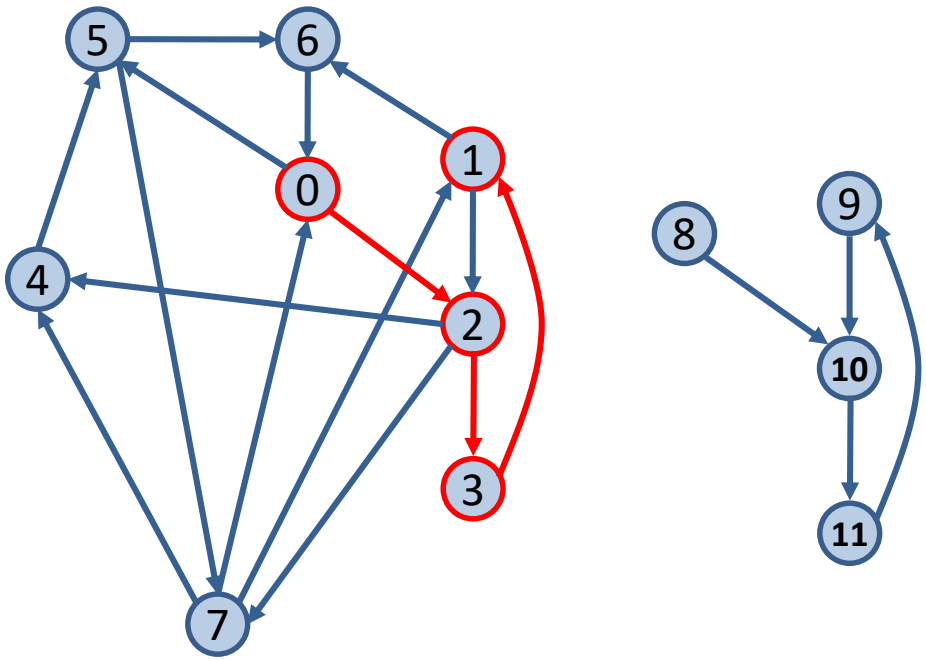


P = {1, 4, 7, 5}

Lv = {0, 2, 3}

Ejemplo de recorrido en profundidad

i = 0

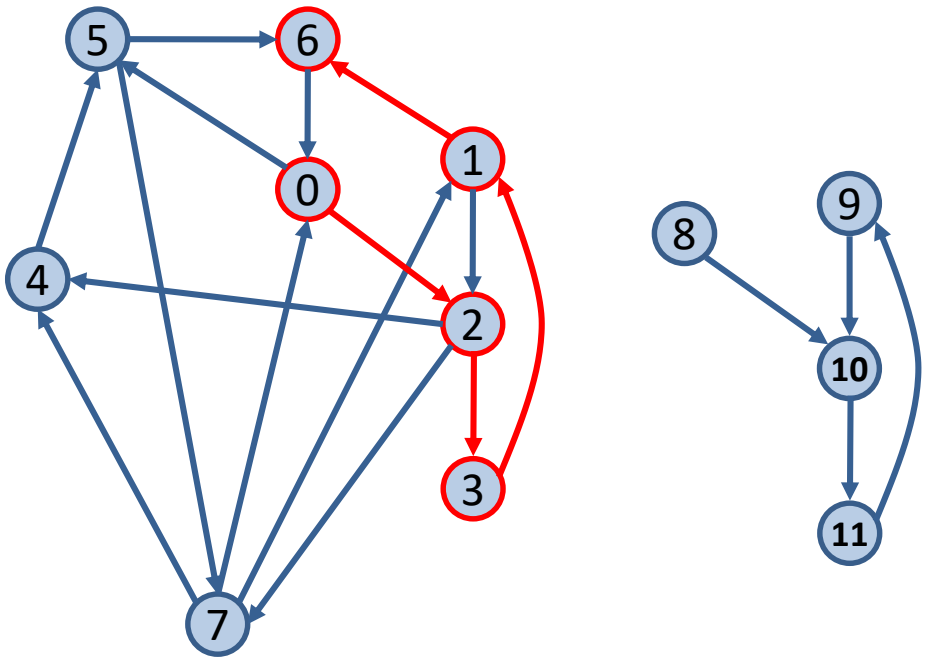


P = {6, 4, 7, 5}

Lv = {0, 2, 3, 1}

Ejemplo de recorrido en profundidad

i = 0

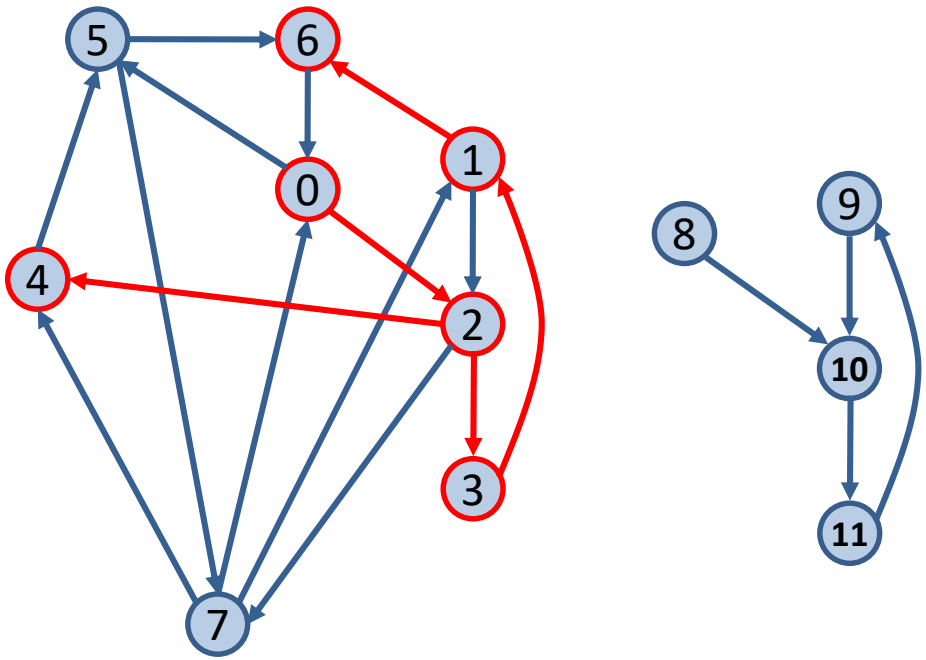


P = {4, 7, 5}

Lv = {0, 2, 3, 1, 6}

Ejemplo de recorrido en profundidad

i = 0

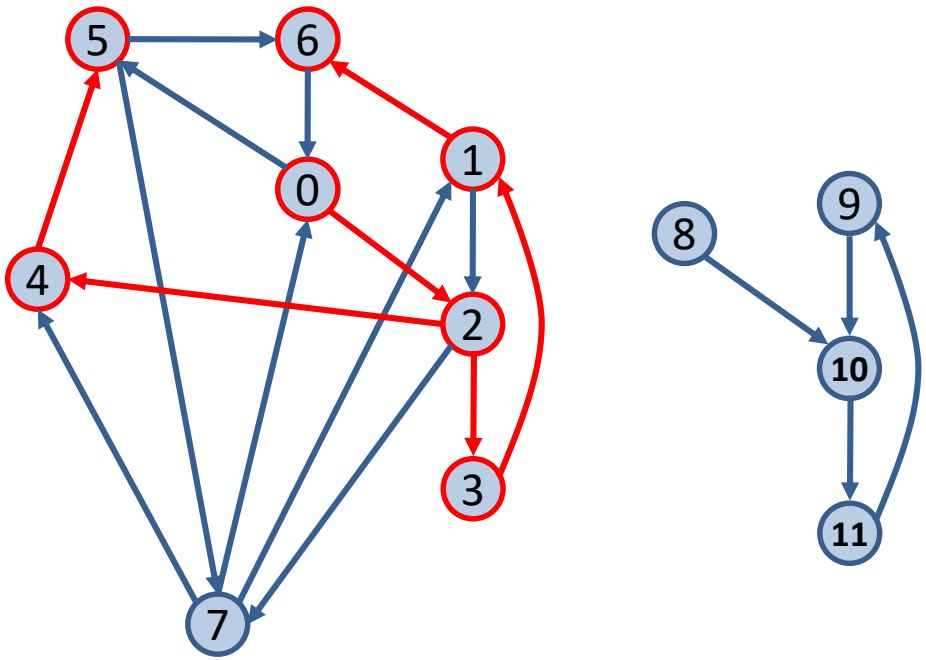


P = {5, 7, 5}

Lv = {0, 2, 3, 1, 6, 4}

Ejemplo de recorrido en profundidad

i = 0

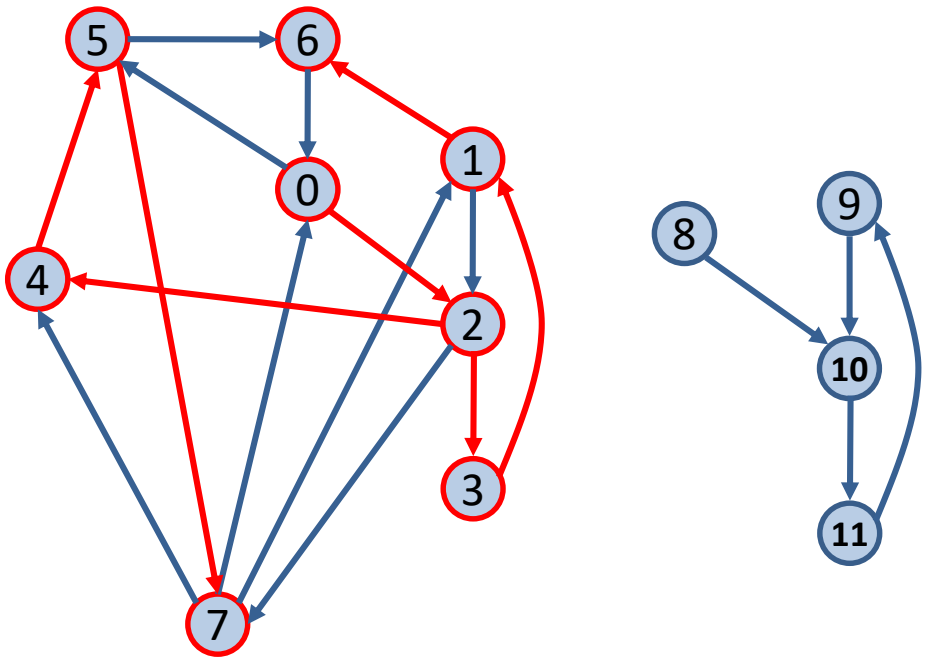


P = {7, 7, 5}

Lv = {0, 2, 3, 1, 6, 4, 5}

Ejemplo de recorrido en profundidad

i = 0

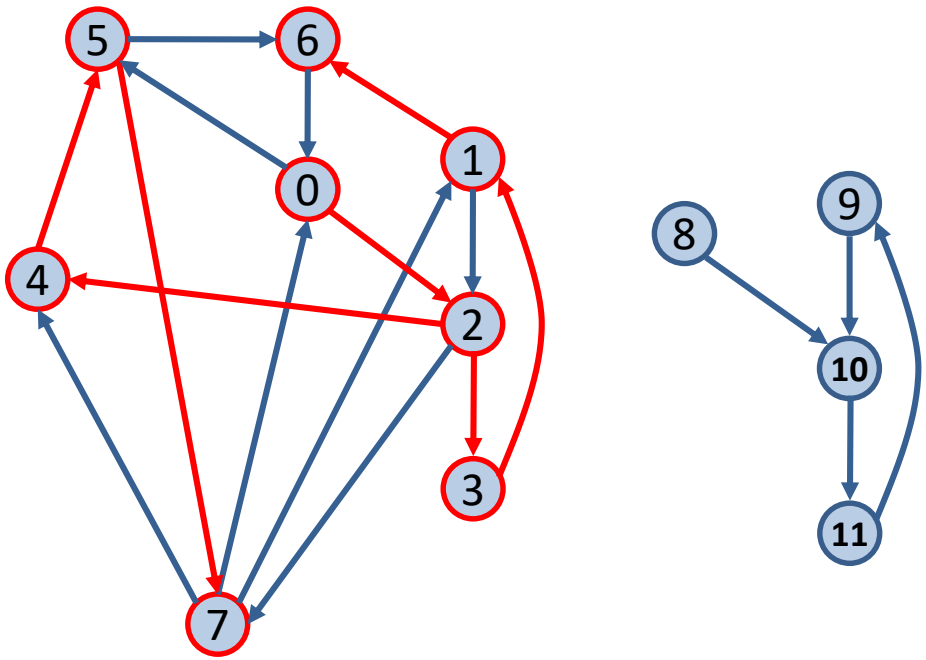


P = {7, 5}

Lv = {0, 2, 3, 1, 6, 4, 5, 7}

Ejemplo de recorrido en profundidad

i = 0



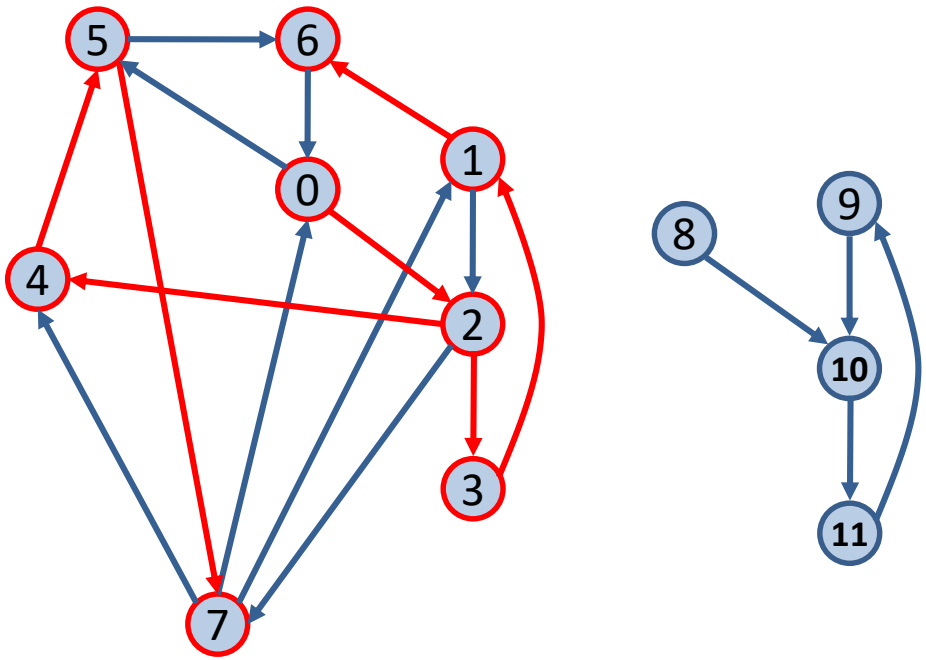
7, VISITADO

P = {5}

Lv = {0, 2, 3, 1, 6, 4, 5, 7}

Ejemplo de recorrido en profundidad

i = 0



5, VISITADO

P = { }

Lv = {0, 2, 3, 1, 6, 4, 5, 7}

Ejemplo de recorrido en profundidad

i = 1

i = 2

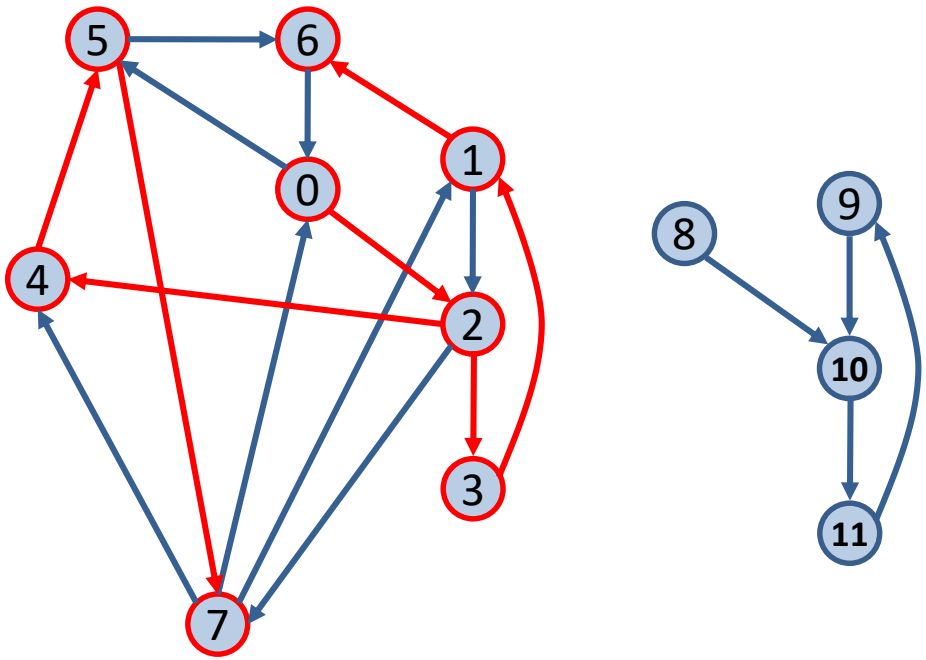
i = 3

i = 4

i = 5

i = 6

i = 7

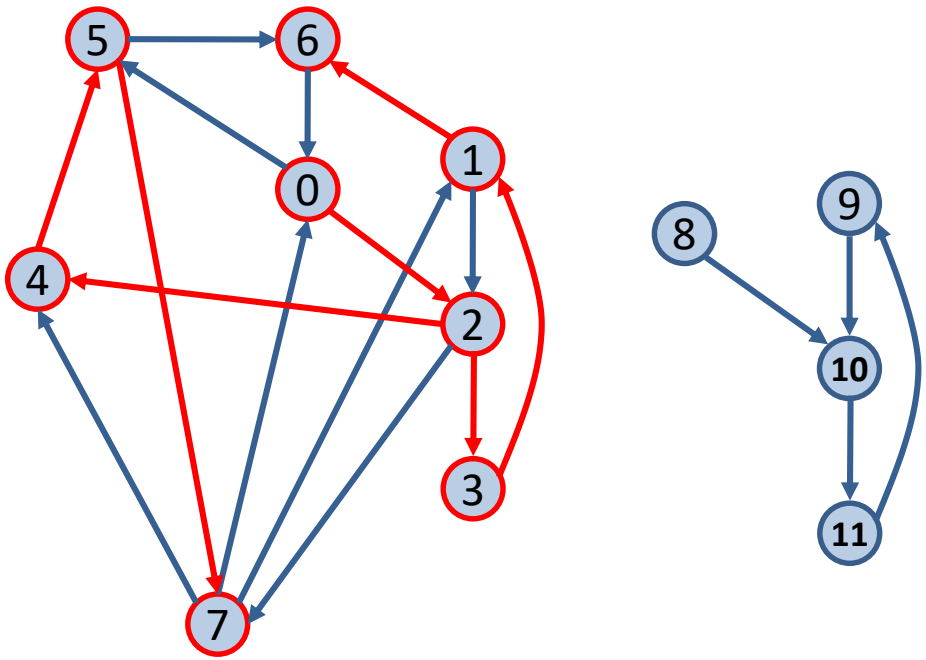


P = { }

Lv = {0, 2, 3, 1, 6, 4, 5, 7}

Ejemplo de recorrido en profundidad

i = 8

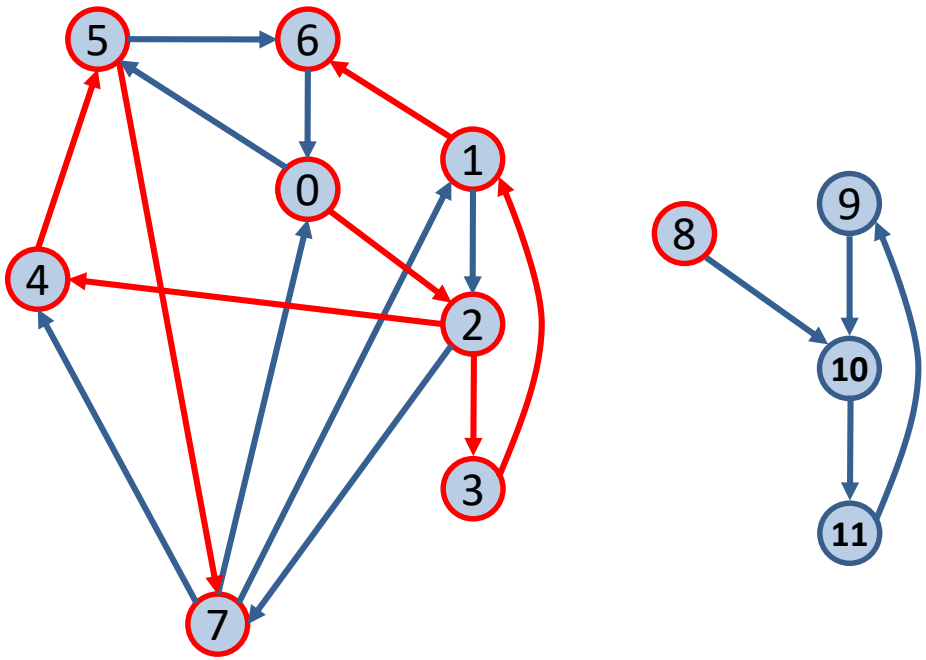


P = {8}

Lv = {0, 2, 3, 1, 6, 4, 5, 7}

Ejemplo de recorrido en profundidad

i = 8

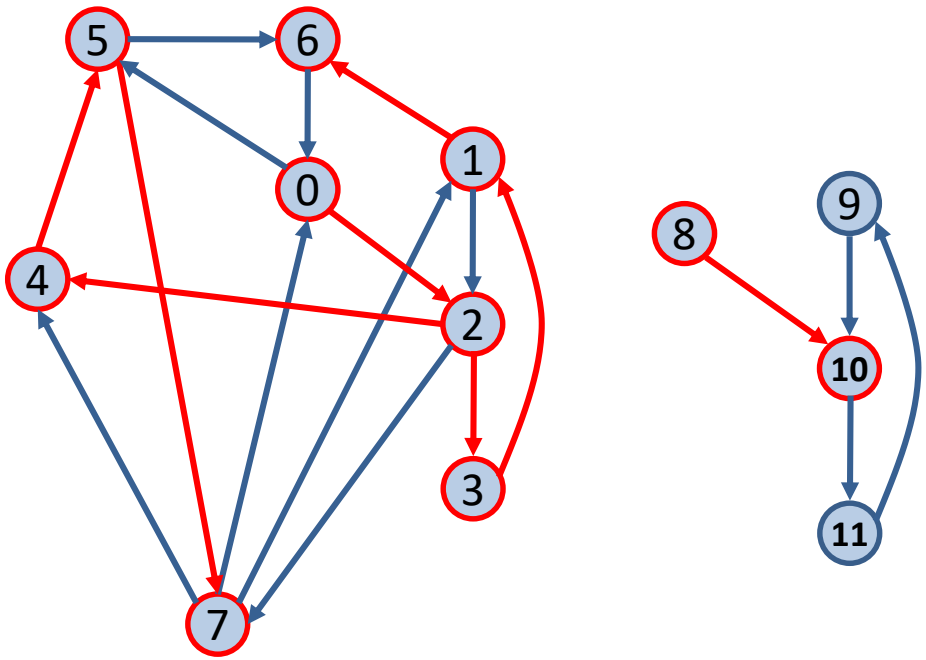


P = {10}

Lv = {0, 2, 3, 1, 6, 4, 5, 7, 8}

Ejemplo de recorrido en profundidad

i = 8

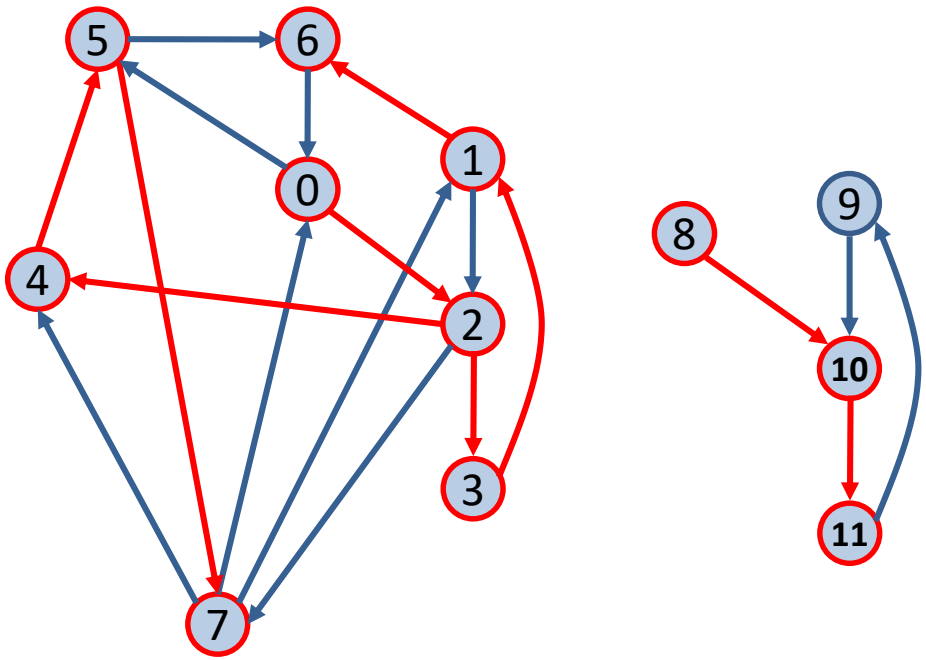


P = {11}

Lv = {0, 2, 3, 1, 6, 4, 5, 7, 8, 10}

Ejemplo de recorrido en profundidad

i = 8

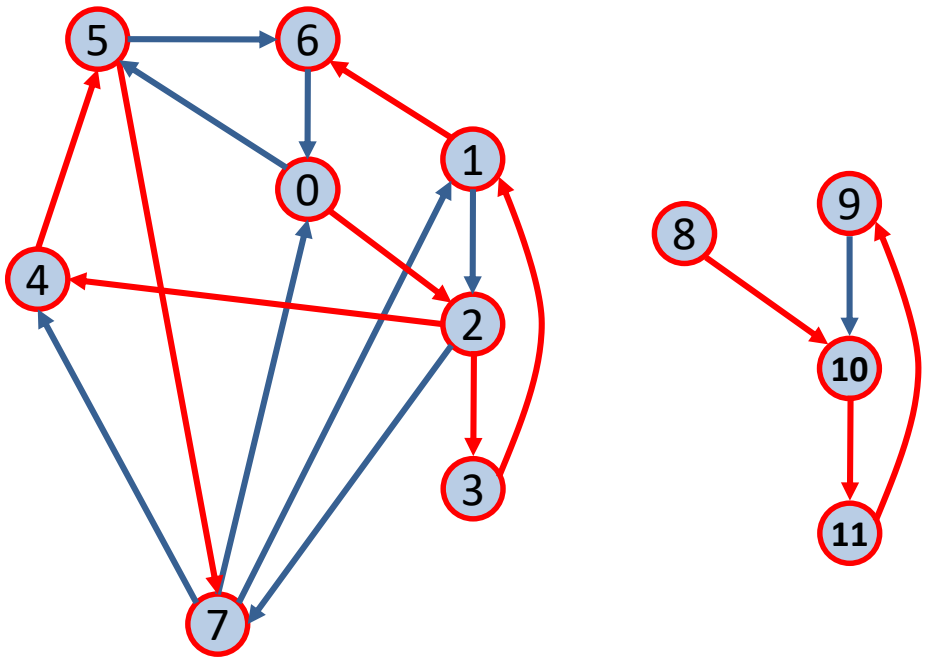


P = {9}

Lv = {0, 2, 3, 1, 6, 4, 5, 7, 8, 10, 11}

Ejemplo de recorrido en profundidad

i = 8



P = { }

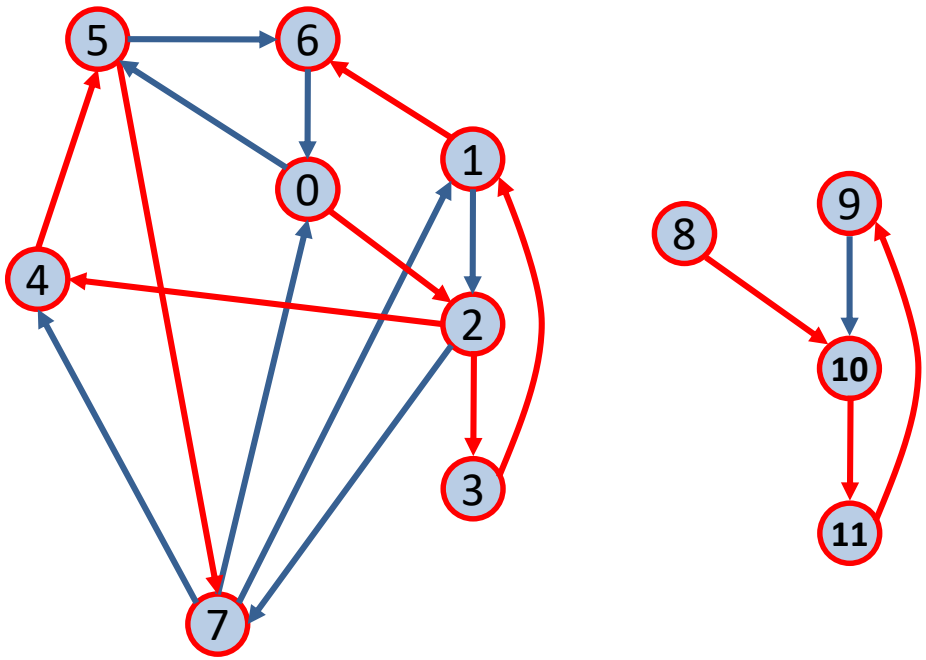
Lv = {0, 2, 3, 1, 6, 4, 5, 7, 8, 10, 11, 9}

Ejemplo de recorrido en profundidad

i = 9

i = 10

i = 11



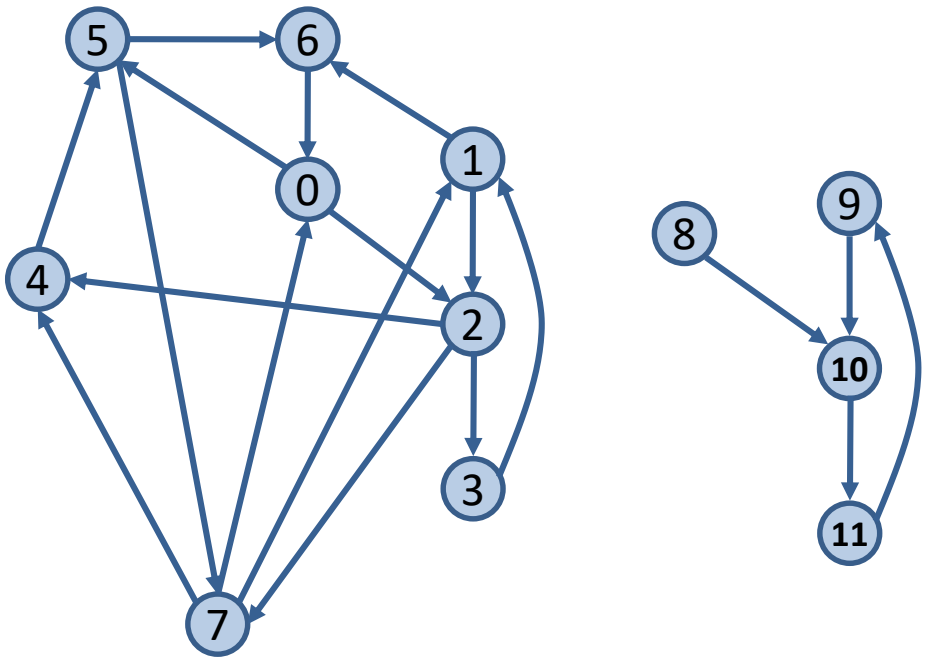
P = { }

Lv = {0, 2, 3, 1, 6, 4, 5, 7, 8, 10, 11, 9}

```
#include "listaenla.h"
#include "colaenla.h"
Lista<Grafo::vertice> Anchura(const Grafo& G, Grafo::vertice u)
{
   typedef Grafo::vertice vertice;
   const size_t n = G.numVert();
   vector<visitas> marcas(n, NO_VISITADO);
   Cola<vertice> C;
   Lista<vertice> Lv;
   vertice i = u;
   do {
      if (marcas[i] == NO_VISITADO) {
         C.push(i);
         do {
            vertice v = C.frente(); C.pop();
            if (marcas[v] == NO_VISITADO) {
               marcas[v] = VISITADO;
               Lv.insertar(v, Lv.fin());
               // Meter en la cola los adyacentes no visitados
               for (vertice w = 0; w < n; ++w)
                  if (G[v][w] && marcas[w] == NO_VISITADO)
                     C.push(w);
            }
         } while (!C.vacia());
      }
      i = (i+1) % n;
   } while (i != u);
   return Lv;
}
```

Ejemplo de recorrido en anchura

i = 0

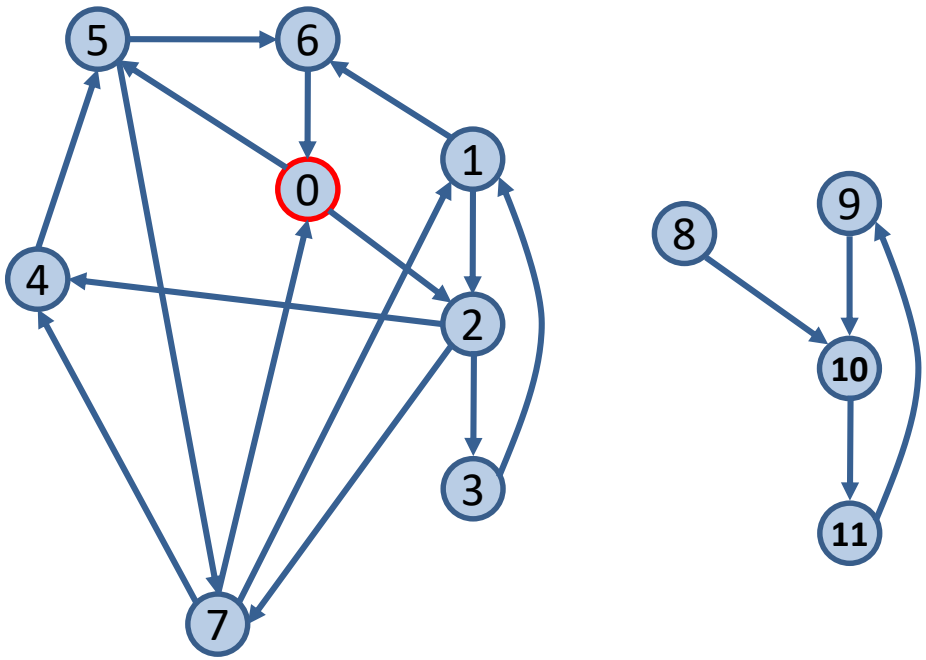


C = {0}

Lv = { }

Ejemplo de recorrido en anchura

i = 0

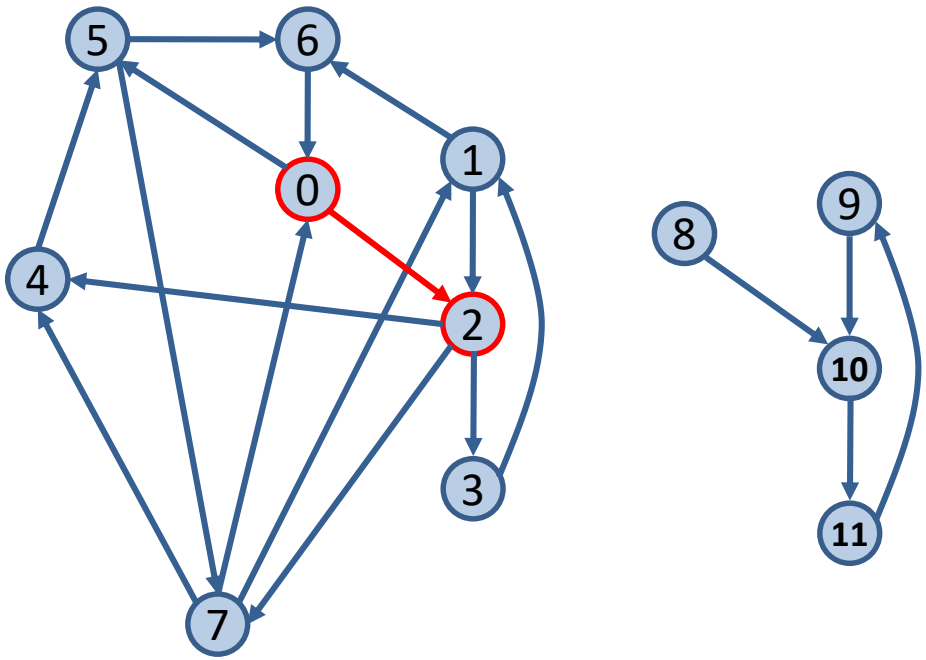


C = {2, 5}

Lv = {0}

Ejemplo de recorrido en anchura

i = 0

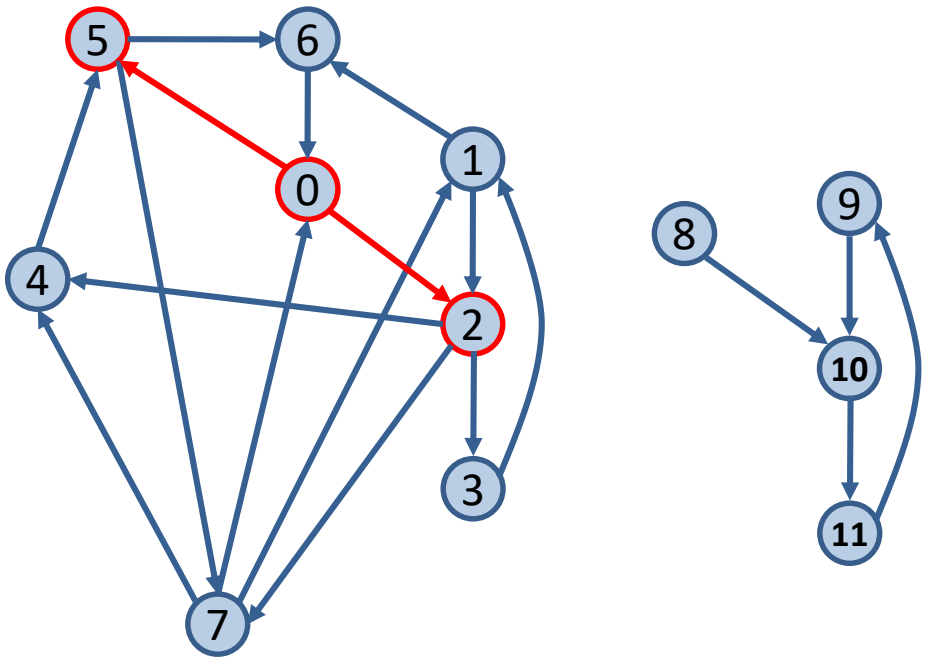


C = {5, 3, 4, 7}

Lv = {0, 2}

Ejemplo de recorrido en anchura

i = 0

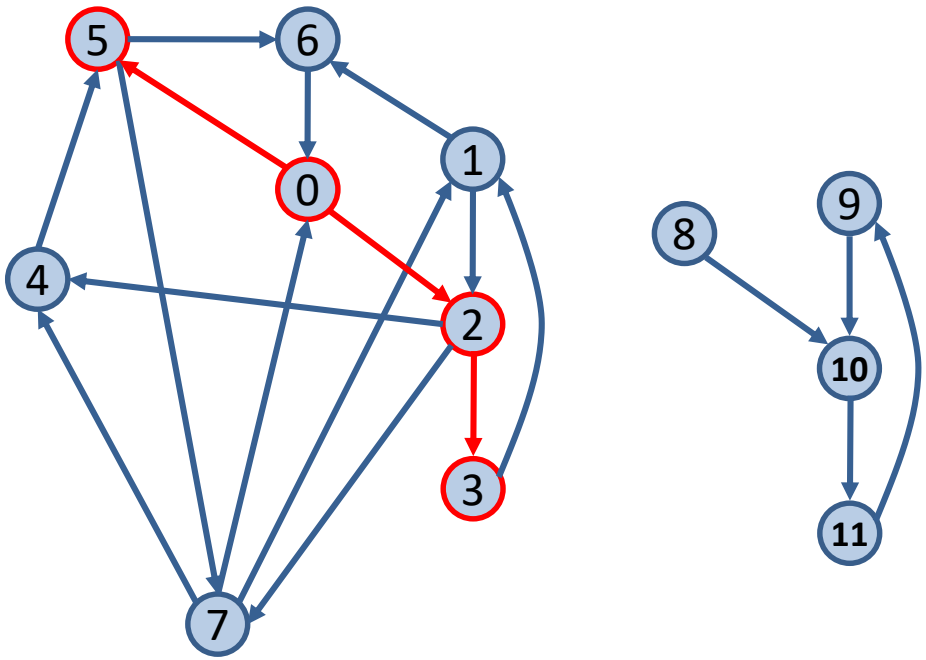


C = {3, 4, 7, 6, 7}

Lv = {0, 2, 5}

Ejemplo de recorrido en anchura

i = 0

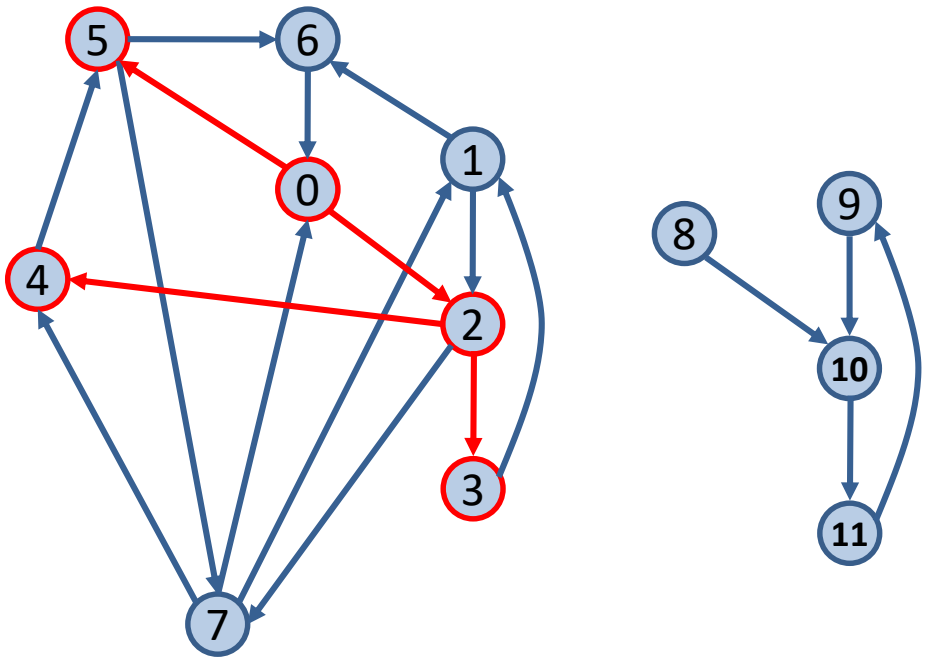


C = {4, 7, 6, 7, 1}

Lv = {0, 2, 5, 3}

Ejemplo de recorrido en anchura

i = 0

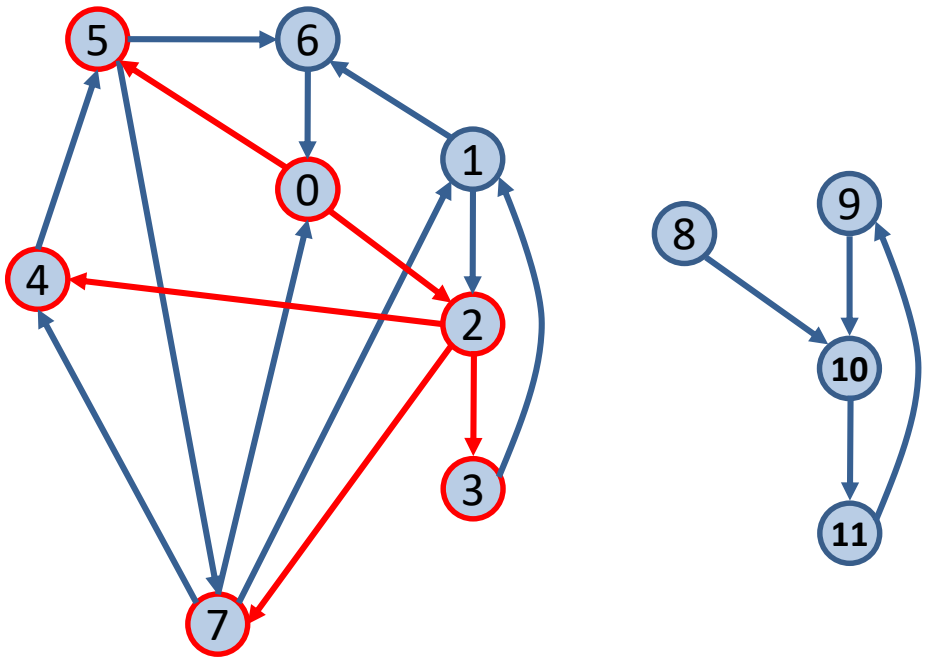


C = {7, 6, 7, 1}

Lv = {0, 2, 5, 3, 4}

Ejemplo de recorrido en anchura

i = 0

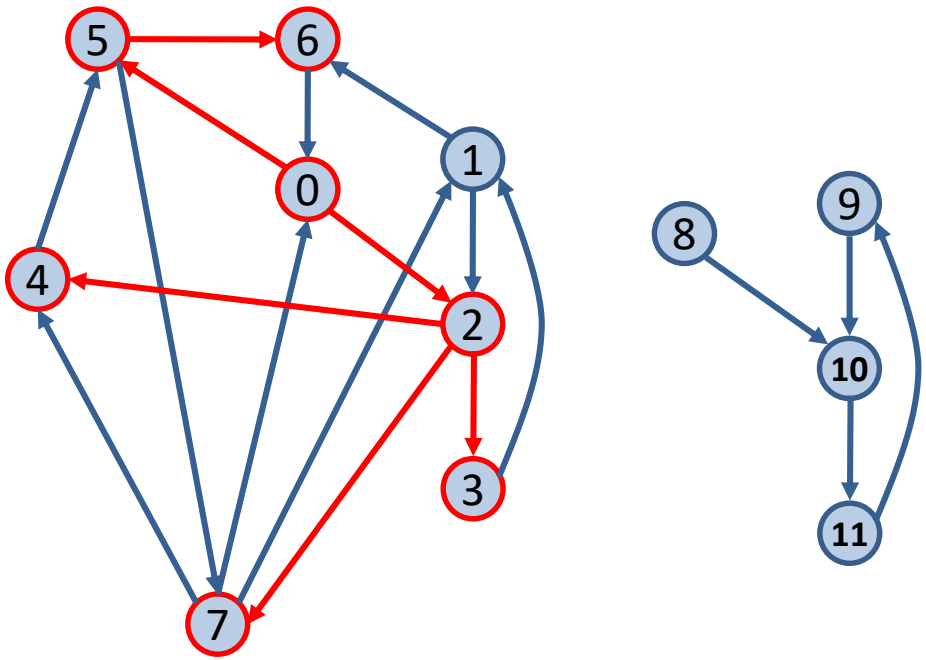


C = {6, 7, 1, 1}

Lv = {0, 2, 5, 3, 4, 7}

Ejemplo de recorrido en anchura

i = 0

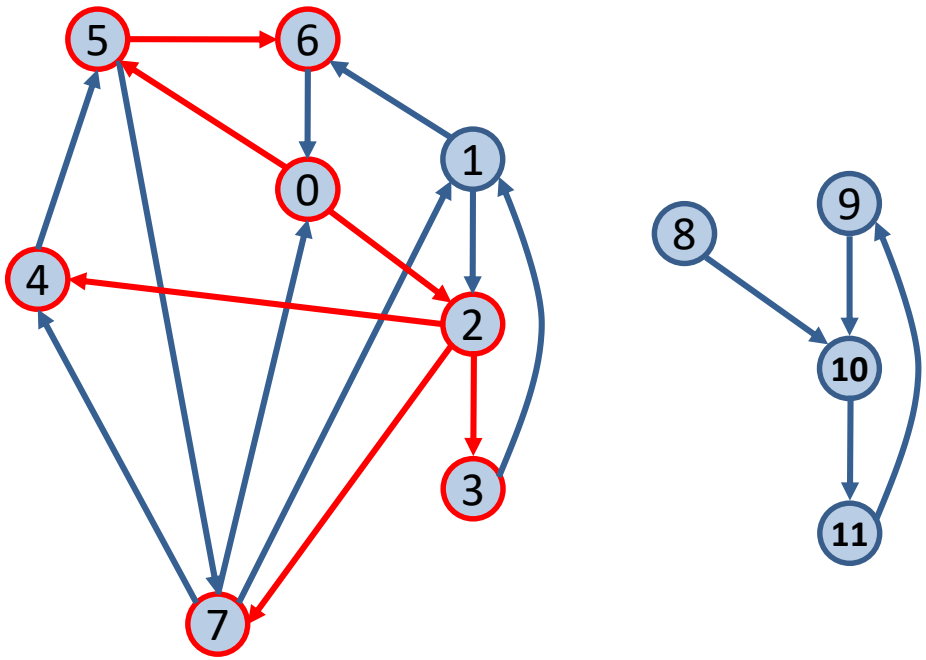


C = {7, 1, 1}

Lv = {0, 2, 5, 3, 4, 7, 6}

Ejemplo de recorrido en anchura

i = 0



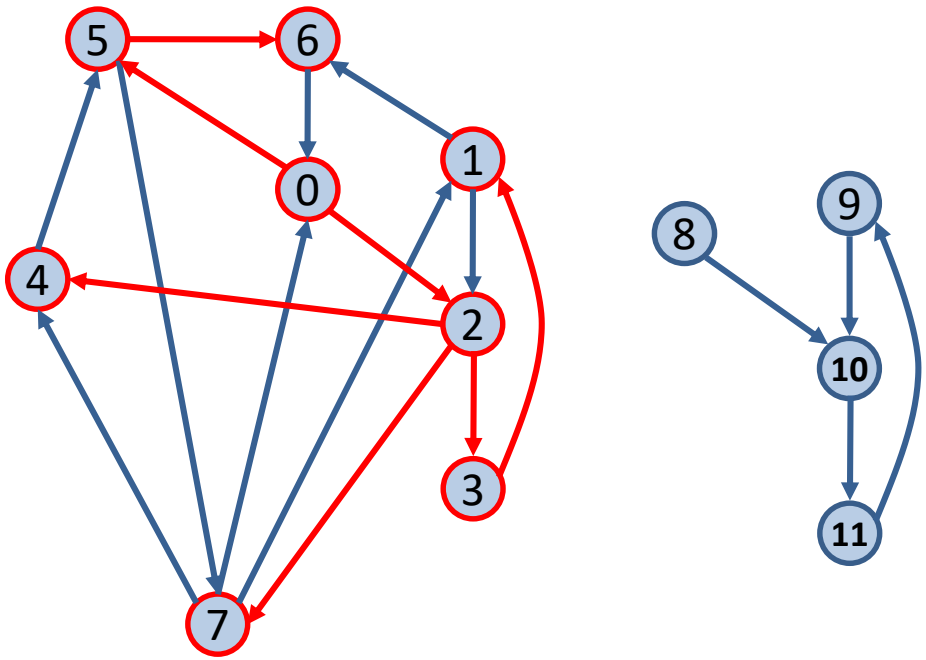
7, VISITADO

C = {1, 1}

Lv = {0, 2, 5, 3, 4, 7, 6}

Ejemplo de recorrido en anchura

i = 0

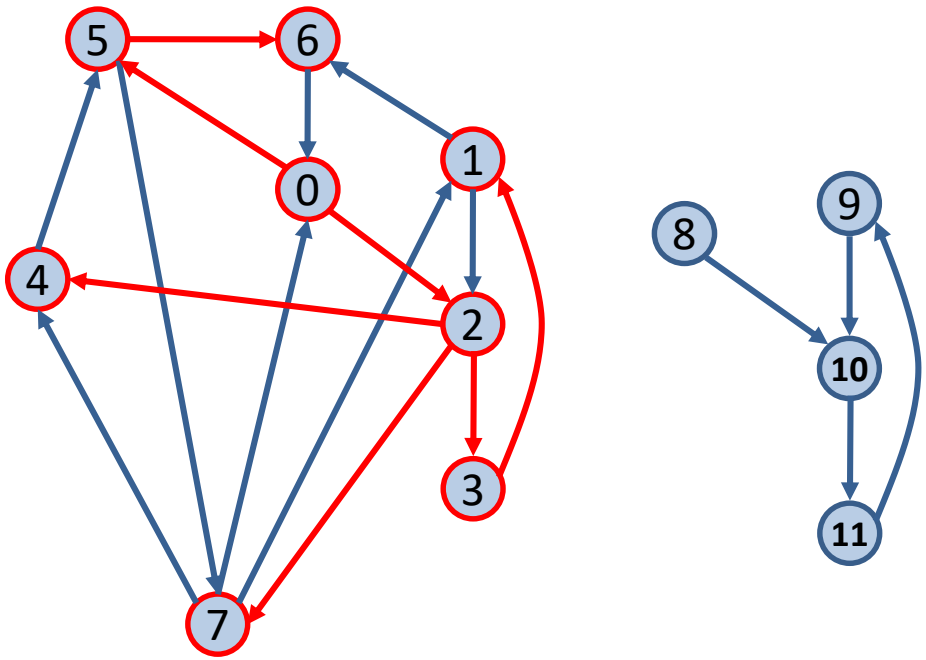


C = {1}

Lv = {0, 2, 5, 3, 4, 7, 6, 1}

Ejemplo de recorrido en anchura

i = 0



1, VISITADO

C = { }

Lv = {0, 2, 5, 3, 4, 7, 6, 1}

Ejemplo de recorrido en anchura

i = 1

i = 2

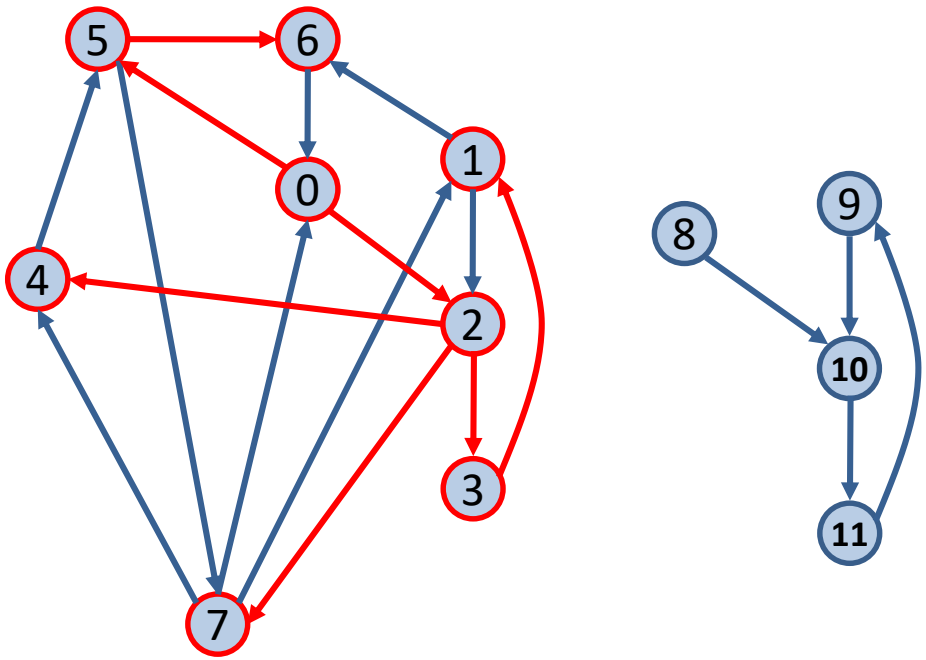
i = 3

i = 4

i = 5

i = 6

i = 7

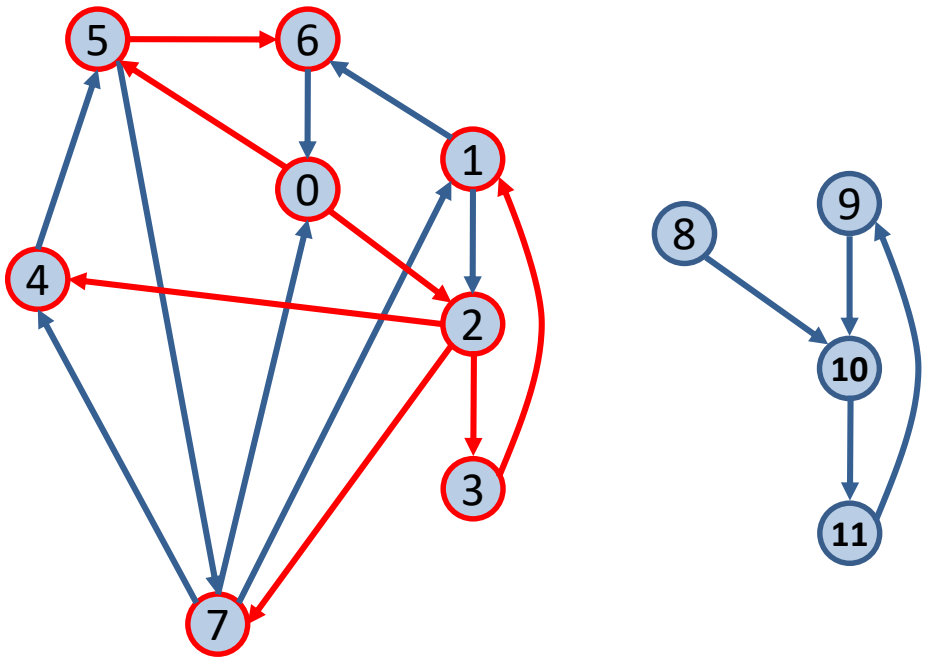


C = { }

Lv = {0, 2, 5, 3, 4, 7, 6, 1}

Ejemplo de recorrido en anchura

i = 8

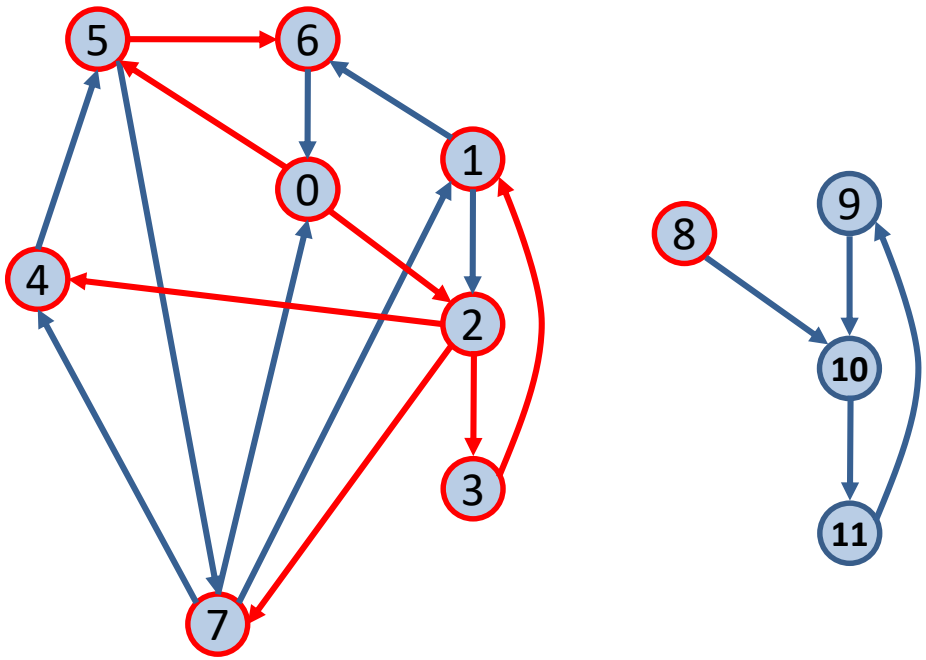


C = {8}

Lv = {0, 2, 5, 3, 4, 7, 1, 6}

Ejemplo de recorrido en anchura

i = 8

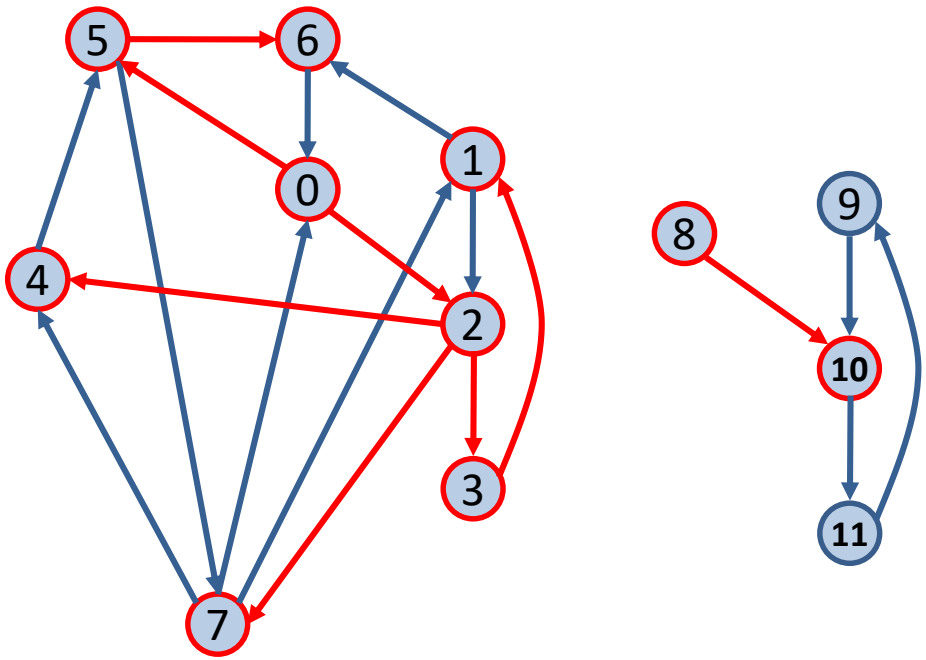


C = {10}

Lv = {0, 2, 5, 3, 4, 7, 1, 6, 8}

Ejemplo de recorrido en anchura

i = 8

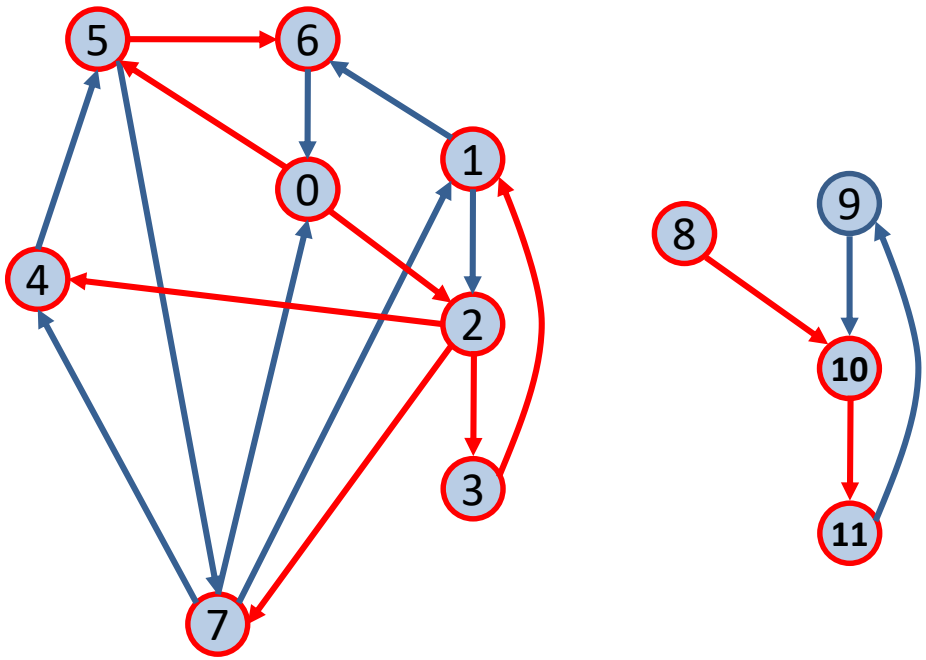


C = {11}

Lv = {0, 2, 5, 3, 4, 7, 1, 6, 8, 10}

Ejemplo de recorrido en anchura

i = 8

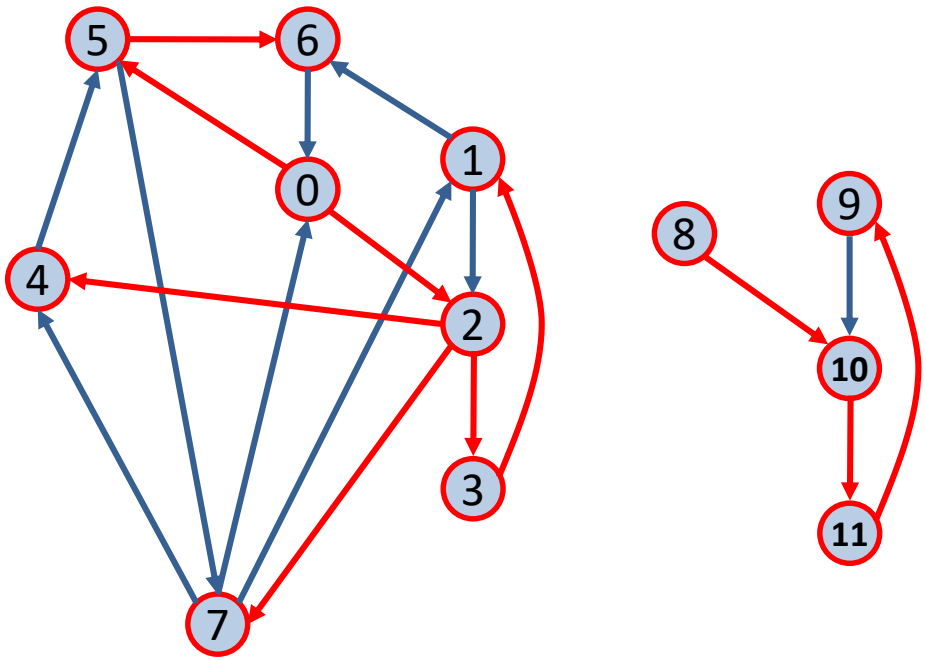


C = {9}

Lv = {0, 2, 5, 3, 4, 7, 1, 6, 8, 10, 11}

Ejemplo de recorrido en anchura

i = 8



C = { }

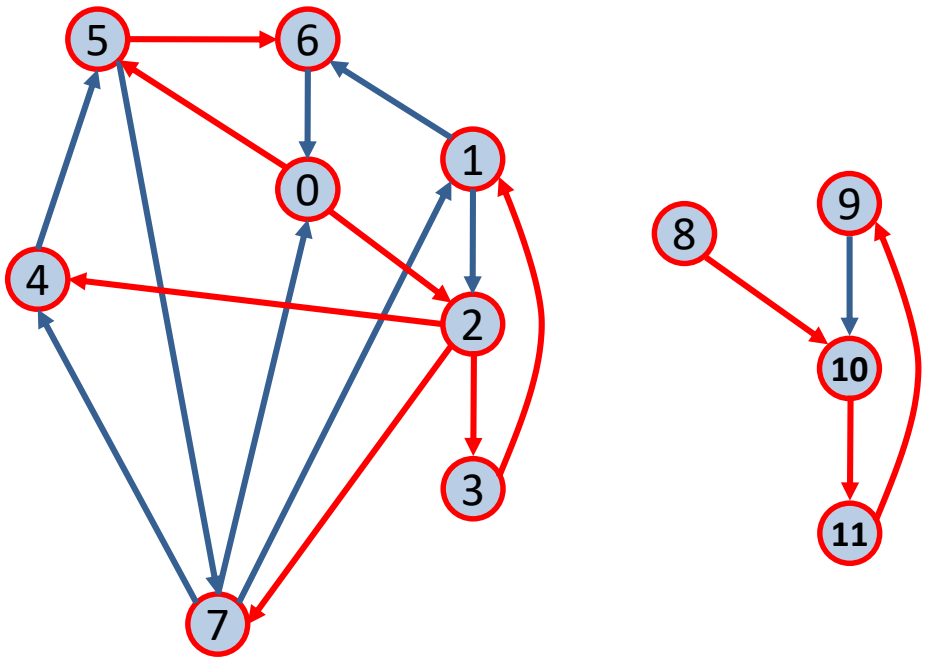
Lv = {0, 2, 5, 3, 4, 7, 1, 6, 8, 10, 11, 9}

Ejemplo de recorrido en anchura

i = 9

i = 10

i = 11



C = { }

Lv = {0, 2, 5, 3, 4, 7, 1, 6, 8, 10, 11, 9}